

Nicolas Ha

Place of birth: Nantes, France

Nationality: French, British

CONTACT

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ABOUT ME

Senior Software Engineer with expert knowledge of Typescript, Clojure, message queues. Comfortable with secure distributed systems in regulated environments. Built banking-grade auth/authz systems and resilient infrastructure processing millions in transactions. Experience with web services at scale (1.3B visitors/month).

DIGITAL SKILLS

JavaScript | TypeScript | Clojure | React.js | Node.js | Kubernetes | Docker

DATABASES

PostgreSQL | Redis | FoundationDB | ElasticSearch | MongoDB

MESSAGE QUEUES

Apache Pulsar | Apache Kafka | RabbitMQ

ENGINEERING

Testing (unit, integration, e2e, generative) | Chaos Engineering

DEVOPS

CI/CD | DataDog | AWS

WORK EXPERIENCE

2022 – CURRENT London, United Kingdom

Senior software engineer Griffin Bank (BaaS)

Joined before the application for the banking licence. Engineered critical platform components essential for banking licence approval in a highly regulated financial environment. All built in-house from scratch to avoid dependencies on 3rd-party systems:

- Implemented secure admin authorization systems for manual operations
- Built the internal login system and multi-factor authentication using passkeys
- Designed resilient queuing and processing systems (influenced by blockchain ideas, Amazon Dynamo and Apple QuiCK papers)
- Created comprehensive audit logging necessary to obtain a banking licence
- Established reliable scheduled job infrastructure
- Implemented secure HTTP message signatures for API integrity from scratch (RFC 9421)
- The security features (2FA, HTTP Message signatures) passed independent security audits
- Kubernetes, Datomic, Kafka, FoundationDB, Bazel

2019 – 2022 London, United Kingdom

Independent Developer TurtleQueue (SaaS)

Led development of TurtleQueue, a Pulsar-compatible message broker SaaS platform.

- Designed and built entire technology stack from scratch
- Gained expertise in distributed systems and multi-tenant cloud infrastructure
- Wrote documentation, published SDKs on npm and maven
- Successfully secured R&D tax incentives
- Joined YC Startup School
- Provided specialized freelance engineering services for MailOnline
- Used Typescript, Terraform, Kubernetes, PostgreSQL, GRPC, GraphQL, Clojure, Docusaurus

2016 – 2019 London, United Kingdom

Backend team lead software engineer MailOnline (Media)

Initially part of the frontend team, quickly moved to the backend team. Advanced from Senior Engineer to Backend Team Lead managing highly experienced developers.

- Evolved from execution-focused engineer to architectural steward preserving system integrity
- Used server-side rendering to allow caching and scaling the core publishing pipeline serving 1.3B views monthly
- Responsible for the comments system (moderation and notifications)
- Implemented ad placement that increased revenue
- Delivered real-time analytics that significantly reduced feedback cycles, accelerating feature delivery
- Used Node.js, Redis, ElasticSearch, RabbitMQ, Zookeeper, Varnish, Akamai, Clojure, BigQuery at scale

2015 – 2016 France

Software engineer Go Find It (Startup mobile apps)

First backend engineer hire.

- Built a full-stack mobile CMS for luxury hotels that enabled real-time content management and live app previewing
- Managed the entire pipeline from development to DevOps and App Store publishing in a team of two
- Supported team of 10 mobile developers to deliver hundreds of branded hotel apps
- Node.js, React.js

2014 – 2014 France

Software engineer Apizee (Startup web telecommunications)

First backend engineer hire.

- Proactively developed a prototype that secured a multi-year contract with Orange
- Developed connectors bridging WebRTC web customers with legacy phone support systems
- Technology showcased at Barcelona Mobile World Congress on HP's booth
- Node.js, Socket.io, WebRTC

2012 – 2012 France

Intern Siléane (Robotics SMB)

Developed intuitive robotic arm controller using Microsoft Kinect and C#.

- Enabled non-technical users to program movement paths through natural gestures
- Reduced programming time compared to traditional joint-by-joint configuration methods

2010 – 2010 France

Intern Dassault Aviation (equipments division)

Developed an automated debugging bench to stress test military avionics flight control circuit boards, enabling 24-hour testing cycles that reduced testing time by 50%.

EDUCATION AND TRAINING

2011 – 2014 Lannion, France

Master's degree ENSSAT

Student representative. Erasmus exchange in Germany, with classes taught in German.

2010 – 2011 Edinburgh, United Kingdom

Bachelor of Engineering in Electronic and Computer Engineering Edinburgh Napier University

LANGUAGE SKILLS

MOTHER TONGUE(S): French

OTHER LANGUAGE(S): English